

PDD_STAT — User Guide

Version: 1.0

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Interface language: English (labels, metrics, plot annotations)

Table of Contents

1. [System Overview](#)
2. [Installation & Startup](#)
3. [Project & Data Management](#)
 4. 3.1. Creating a project and uploading data
 5. 3.2. Data cleaning
 6. 3.3. Variable management
 7. 3.4. Project context (description, aim)
8. [Interface](#)
 9. 4.1. Left panel — Project Manager
 10. 4.2. Workspace
 11. 4.3. Right panel — Variables
12. [Statistical Analysis Templates](#)
 13. 5.1. Categorical Comparison
 14. 5.2. Numeric Comparison
 15. 5.3. Correlation Analysis
 16. 5.4. Spline Analysis
 17. 5.5. Descriptive Statistics
 18. 5.6. Violin Plots
 19. 5.7. ANOVA / Kruskal-Wallis
 20. 5.8. Chart Builder
 21. 5.9. Cox Regression
 22. 5.10. Logistic Regression
 23. 5.11. Logistic LASSO

24.	5.12.	Random Forest
25.	5.13.	Kaplan-Meier
26.	5.14.	ROC Analysis
27.	5.15.	Model Evaluation (Binary)
28.	5.16.	Survival Prediction Evaluation
29.	5.17.	Diagnostic Accuracy
30.	5.18.	Agreement Analysis
31.	5.19.	Individual Prediction
32.		AI Chat
33.	6.1.	AI Settings
34.	6.2.	Modes: Assistant and Coder
35.		Reports
36.	7.1.	Standard DOCX Report
37.	7.2.	AI Report
38.		Code Editor Mode
39.		Analysis History (Chain)
40.		Tips & Best Practices

1. System Overview

PDD_STAT is a web application for statistical analysis of medical and clinical data. The system provides:

- **22 analysis templates** (from descriptive statistics to regression models and survival analysis)
- **AI assistant** powered by Ollama (local) or Groq (cloud API)
- **DOCX report generation** (standard and AI-generated)
- **Code editor** for custom Python analysis
- **Visualizations**: matplotlib, SHAP, DCA, calibration curves

Key Features

- **All output in English** (metrics, tables, plot labels)
- **Built-in statistical safeguards**: proportional hazards testing (Schoenfeld), multicollinearity (VIF), normality tests, homogeneity of variances, calibration (Hosmer-Lemeshow)
- **Multiple comparison correction** (Bonferroni) in pairwise tests
- **Stratified survival analysis**
- **Stepwise predictor selection** with p-value thresholds

- **DCA** (Decision Curve Analysis) and **calibration curves** for regression models
- **Bootstrap** confidence intervals for metrics
- **SHAP** for Random Forest model interpretation

2. Installation & Startup

Requirements

Component	Version
Python	≥ 3.11
pip	≥ 21.0

Quick Start (one-click)

macOS:

```
./setup_mac.command && ./start_backend.command
```

Windows:

```
setup.bat  
start.bat
```

The scripts auto-detect Python 3.11+, create a virtual environment, install dependencies, and start the server.

Manual Setup (alternative)

```
cd app/backend  
pip install -r requirements.txt  
python -m uvicorn app.main:app --host 127.0.0.1  
--port 8000
```

Open your browser at: **<http://127.0.0.1:8000>**

3. Project & Data Management

3.1. Creating a Project and Uploading Data

1. Click "+ Create Project" (left panel).
2. Enter a project name (Latin or Cyrillic).
3. Select a data file:
4. Supported formats: `.xlsx`, `.xls`, `.csv`
5. Click "Create".
6. After upload, the **Cleaning Recommendations** modal appears (see section 3.2).

Re-uploading to an existing project: click the project name with the left mouse button — a file dialog opens.

3.2. Data Cleaning

Upon data upload, the system automatically analyzes every column and suggests a cleaning plan:

Dropped Columns

Rule	Condition	Level
HIGH_MISSING	>30% missing values	Hard
MODERATE_MISSING	20–30% missing values	Soft
CONSTANT	Singleton column	Hard
IDENTIFIER	All unique values (n>10)	Hard
TECHNICAL	Name starts with <code>INFO_</code> , <code>CHECK_</code> , <code>PARAMS_</code> etc.	Hard

Imputation

Remaining columns with missing values: - **Numeric** → filled with **median** - **Categorical/string** → filled with **mode**

Buttons: - "Apply Cleaning" — apply recommendations - "Skip (Keep as is)" — save raw data as-is

After cleaning, data is persisted as `state/project_data.parquet` (snappy compression).

3.3. Variable Management

Right panel → "Columns" tab:

- **Search** by variable name
- **Type badge:** `num` (numeric), `cat` (categorical), `bin` (binary), `str` (string)
- **Actions:**
 -  — rename
 -  — delete from dataset
 -  — open Label modal:
 - **Chart Name** — display name on plots
 - **Value Labels** — label for values (e.g., `0 = No`, `1 = Yes`)

Quick variable selection: click a variable name to assign it to the active card field (field highlights in color).

3.4. Project Context (Description, Aim)

Click  on a project item → **Project Context:**

- **Description** — project description (used by AI for context)
- **Research Aim** — study objective
- **Notes** — additional notes

Context is automatically injected into the AI assistant's system prompt and AI reports.

4. Interface

4.1. Left Panel — Project Manager

Element	Description
+ Create Project	Create a new project
Project list	Project names. Click → open/upload data.  → context. × → delete
AI Settings	Configure AI assistant
◀ ▶	Collapse/expand panel

4.2. Workspace

Central area — container for analysis cards (see section 5).

Template menu bar (top):

Menu	Templates
Basic Stats ▼	Categorical, Numeric, Correlation, Spline, Descriptive, Violin, ANOVA, Chart Builder
Regression ▼	Cox, Logistic, LASSO, Random Forest
Survival ▼	Kaplan-Meier
Evaluation ▼	Model Evaluation (Binary), Survival Eval, ROC, Diagnostic Accuracy, Agreement
Prediction ▼	Individual Prediction

Action buttons: - **AI Chat** — open AI assistant chat - **Code** — switch to code editor mode - ☀ / ☾ — toggle light/dark theme

4.3. Right Panel — Variables

Tab	Content
Columns	Variable list with search, type badges, editing
Reports	DOCX report generation
Chain	Analysis history

5. Statistical Analysis Templates

5.1. Categorical Comparison

Method: χ^2 -test / Fisher's Exact Test / Monte Carlo simulation

Parameters: | Field | Description | |-----|-----| | `col1` | First categorical variable (table rows) | | `col2` | Second categorical variable (table columns) |

Test selection logic: 1. All expected frequencies $\geq 5 \rightarrow \chi^2$ -test
2. Low expected AND 2x2 table $\rightarrow$ **Fisher's Exact Test** 3. Otherwise $\rightarrow$ **Monte Carlo** (10,000 iterations)

Output metrics:

Metric	Interpretation
<code>chi2</code>	Chi-square statistic
<code>p_value</code>	Significance level; p < 0.05 = significant association
<code>df</code>	Degrees of freedom
<code>min_expected_frequency</code>	Minimum expected cell count
<code>all_expected_ge5</code>	All expected ≥ 5 ?
<code>odds_ratio</code>	Odds ratio (Fisher, 2x2 only)

5.2. Numeric Comparison

Method: t-test (independent) / ANOVA or Mann-Whitney U / Kruskal-Wallis H

Parameters: | Field | Description | |-----|-----| | value | Numeric dependent variable | | group | Categorical grouping variable |

Test selection: - If **normal** (Shapiro-Wilk) AND **equal variances** (Levene): - 2 groups → **t-test** + Cohen's d - >2 groups → **ANOVA** + η^2 - Otherwise: - 2 groups → **Mann-Whitney U** - >2 groups → **Kruskal-Wallis H**

Output metrics:

Metric	Interpretation
statistic	Test statistic value
p_value	Significance level
cohens_d	Cohen's d effect size (t-test only): 0.2 = small, 0.5 = medium, 0.8 = large
normality	Shapiro-Wilk test results per group
equal_var	Levene's test result

Plots: Boxplot + barplot (mean $\pm$ SD), 2 panels.

5.3. Correlation Analysis

Method: Pearson (parametric) or Spearman (rank) correlation

Parameters: | Field | Description | |-----|-----| | predictors | Select multiple variables | | method | Pearson (assumes normality) or Spearman (any distribution) | | threshold | | cutoff: 0.3, 0.5 (default), 0.7 |

Output metrics:

Metric	Interpretation
n_variables	Number of variables
n_strong_pairs	Pairs with
strongest_r	Maximum
pairs	Strong pairs with Bonferroni-corrected p-values . Corrects for family-wise error rate (FWER)

p-values are Bonferroni-corrected (multiplied by number of all pairwise tests) to control FWER.

Plot: Correlation heatmap (lower triangle).

5.4. Spline Analysis

Method: Linear spline analysis within Cox or Logistic regression

Parameters: | Field | Description | |-----|-----| |
 variable | Variable for spline transformation | | target |
 (Logistic) Binary outcome | | time + event | (Cox) Time-to-
 event | | covariates | Adjustment covariates | | knots | Knot
 placement: Median , Quartiles , Custom |

Output metrics:

Metric	Interpretation
nonlinear_p	p-value for non-linearity test. p < 0.05 = significant deviation from linearity
knots	Knot positions

Plot: Log-HR / Log-OR vs variable with knot markers + histogram.

5.5. Descriptive Statistics

Method: Summary statistics + Shapiro-Wilk normality test

Parameters: | Field/Option | Description | |-----|-----|
 | `covariates` | Variables (empty = all numeric) | | `Include`
`histograms` | Plot histograms per variable |

Output metrics:

Metric	Interpretation
N, Mean, Median, Std, Min, Max, Q1, Q3	Standard descriptive statistics
Skewness	Asymmetry of distribution (0 = symmetric)
Kurtosis	Tail heaviness (3 = normal distribution)
Missing count	Number of NaN values
Outlier count	IQR method: beyond $Q1 - 1.5 \times IQR$ / $Q3 + 1.5 \times IQR$
Shapiro-Wilk W, p	$p < 0.05$ = distribution deviates from normality

5.6. Violin Plots

Method: Violin plots + Mann-Whitney / Kruskal-Wallis test

Parameters: | Field | Description | |-----|-----| | `value` |
 Numeric variable | | `group` | Grouping variable |

Output metrics:

Metric	Interpretation
<code>mann_whitney_p</code>	Mann-Whitney U p-value (2 groups)
<code>kruskal_p</code>	Kruskal-Wallis H p-value (>2 groups)

Plot: Violin plot, optional box overlay and swarm points.

5.7. ANOVA / Kruskal-Wallis

Method: One-way ANOVA (parametric) or Kruskal-Wallis (non-parametric)

Parameters: | Field | Description | |-----|-----| | value |
 Numeric outcome | | group | Categorical grouping (≥2 groups)
 |

Output metrics (tabs):

Statistics tab:

Metric	Interpretation
test_name	ANOVA or Kruskal-Wallis
F_stat / H_stat	Test statistic
p_value	Significance level
eta_sq	η^2 — proportion of variance explained by group (ANOVA)
omega_sq	ω^2 — less biased effect size than η^2
epsilon_sq	ϵ^2 — effect size for Kruskal-Wallis
trend_test	Jonckheere-Terpstra trend test (≥3 groups)

Post-hoc tab:

Metric	Interpretation
Pairwise comparisons	Tukey HSD (ANOVA) or Mann-Whitney with Bonferroni (KW)
significant	$p < 0.05$ after correction

Diagnostics tab:

Metric	Interpretation
all_normal	All groups normal (Shapiro-Wilk $p > 0.05$)
equal_var	Variances equal (Levene $p > 0.05$)

Plot: Boxplot + jitter + barplot (mean ± SD).

5.8. Chart Builder

Method: Quick visualization (no statistical tests)

Parameters: | Field | Description | |-----|-----| | Chart type | Histogram, Box Plot, Scatter, Bar (mean±SD), Density | | x | X variable | | y | Y variable (scatter only) | | group | Color/group by | | Bins | Histogram bins (default 20) | | KDE curve | Overlay kernel density estimate (histogram) | | Regression line | Regression line with r^2 (scatter) | | Chart Axes | Custom title, axis labels |

Plot: The requested chart type.

5.9. Cox Regression

Method: Cox Proportional Hazards Regression (lifelines CoxPHFitter)

Parameters: | Field | Description | |-----|-----| | event | Event indicator (0 = censored, 1 = event) | | time | Time-to-event | | covariates | Predictors | | Covariate type | Cat / Cont toggle — categorical or continuous | | Reference groups | Reference category for categorical variables | | **Univariate** | Each predictor separately | | **Multivariate** | All predictors jointly | | **Method** | Enter (all at once), Forward (forward selection), Backward (backward elimination) | | p_enter | Entry p-value threshold (default 0.05) | | p_remove | Removal p-value threshold (default 0.10) | | Adjust Survival Curves | Covariate for adjusted survival curves |

Output metrics:

Table tab:

Metric	Interpretation
Beta (β)	Regression coefficient
SE	Standard error
z	z-statistic (β /SE)
HR (Hazard Ratio)	$\exp(\beta)$. >1 = increased hazard, <1 = decreased hazard
95% CI	95% confidence interval for HR
p-value	Wald test significance

Diagnostics tab:

Metric	Interpretation
C-index	Harrell's Concordance Index — discrimination. 0.5 = random, 1.0 = perfect. Not calibrated for censored data
AIC	Akaike Information Criterion — model quality (lower = better). Not interpretable in isolation
BIC	Bayesian Information Criterion — penalizes complexity more than AIC
Log-likelihood	Log-partial likelihood at convergence
Global LRT p	Likelihood ratio test p-value for overall model significance
Schoenfeld test	Proportional hazards assumption test. p<0.05 = PH violation. Critical diagnostic check
VIF	Variance Inflation Factor. >10 = problematic multicollinearity, >5 = moderate

Steps tab (Forward/Backward only): stepwise selection history.

"Save Risk" button — saves predicted risk (linear predictor) as a new column.

Plots (checkboxes, all optional): - **Forest plot** — HR with 95% CI - **Survival curves** — adjusted survival curves - **Residuals** — Martingale residuals (linearity diagnostic)

Assumption checks: - EPV (Events Per Variable) ≥ 10 — warning if <10 - Schoenfeld — PH test per variable and globally - VIF — multicollinearity - **Stepwise warning:** p-values do not account for the selection process and may be inflated

5.10. Logistic Regression

Method: Logistic Regression with L2 penalty,

```
class_weight='balanced'
```

Parameters: | Field | Description | |-----|-----| | target | Binary target variable | | predictors | Predictors | | Covariate type | Cat / Cont toggle | | Reference groups | Reference

category | | **Univariate** | Each predictor separately | | **Multivariate** | All predictors jointly | | **Method** | Enter , Forward , Backward | | **Validation** | None , Train/Test (80/20), Cross-val (5-fold CV) | | **Plots** | Checkboxes: ROC, DCA, Calibration. All optional, select what you need |

Output metrics:

Table tab:

Metric	Interpretation
Beta (β)	Regression coefficient
OR (Odds Ratio)	$\exp(\beta)$. >1 = higher odds, <1 = lower odds
95% CI	95% confidence interval for OR
p-value	Wald test significance

Diagnostics tab:

Metric	Interpretation
AUC	Area Under ROC Curve — discrimination. 0.5 = random, 0.8 = excellent, 1.0 = perfect
Accuracy	$(TP+TN)/(Total)$ — overall correct classification
Sensitivity	$TP/(TP+FN)$ — true positive rate
Specificity	$TN/(TN+FP)$ — true negative rate
Hosmer-Lemeshow	Calibration test: χ^2 , df, p-value. $p < 0.05$ = significant miscalibration
VIF	Multicollinearity (>10 = problematic)
Warnings	VIF errors, convergence issues

Steps tab (Forward/Backward): stepwise selection history.

"Save Probabilities" button — saves predicted probabilities as a new column.

Plots (checkboxes, all optional): - ROC curve - DCA (Decision Curve Analysis) with Treat All / Treat None baselines - Calibration curve with Brier Score

Assumption checks: - Target must be binary (exactly 2 unique values) - Minimum 20 complete cases - All predictors

standardized (StandardScaler) - **Warning** when
validation="none": metrics are optimistic

5.11. Logistic LASSO

Method: L1-regularized logistic regression (LASSO)

Parameters: | Field | Description | |-----|-----| | `target` | Binary target | | `covariates` | Predictors (empty = all numeric + encoded categorical) | | `Auto-select C` | Automatic C selection via 5-fold CV | | `C` | Inverse regularization strength (if auto-C off). Smaller C = stronger regularization | | **Plots** | Checkboxes: Coef Plot, CV Curve, DCA, Calibration. All optional |

Output metrics:

Metric	Interpretation
AUC	ROC AUC (training data — optimistic)
Accuracy	Overall accuracy (training — optimistic)
<code>n_features_selected</code>	Features with non-zero coefficient
<code>n_features_zeroed</code>	Features zeroed out (removed by LASSO)
<code>best_C</code>	Best C value (if auto-select)
<code>selected_features</code>	List: feature, OR, coefficient
<code>intercept</code>	Model intercept

Plots (checkboxes, all optional): - Coefficient bar plot (selected + top 30 all features) - CV curve (if auto-select C) - DCA (Decision Curve Analysis) - Calibration curve

Important: - Metrics are on **training data** — **optimistic** for generalization - `class_weight='balanced'` — handles class imbalance - `solver='saga'` — appropriate for L1 penalty

5.12. Random Forest

Method: Random Forest Classifier (sklearn, 200 trees, max_depth=10, min_samples_leaf=5, class_weight='balanced')

Parameters: | Field | Description | |-----|-----| | target | Target variable (binary or multiclass) | | exclusions | Columns to exclude (comma-separated) | | Top features | 10, 15 (default), 20, All | | SHAP analysis | Enable SHAP calculation (may be slow on large datasets) |

Output metrics:

Metric	Interpretation
OOB score	Out-of-Bag R ² — unbiased performance estimate from unused trees
Accuracy	Test set accuracy (75/25 split)
AUC	ROC AUC (binary targets only)
Precision	PPV = TP/(TP+FP)
Recall	Sensitivity = TP/(TP+FN)
F1	Harmonic mean of precision and recall
top_features	Gini importance — contribution of each feature to node purity. Not a p-value!
shap_features	Mean absolute SHAP — more interpretable feature contribution magnitude

Plots: - Feature importance bar (Gini) - SHAP bar + summary plots (if enabled)

Post-run buttons: - "Save Model" — saves model as .joblib (with metadata: features, target, metrics) - "Save Probabilities" — saves predicted probabilities as new column

Note: Auto-excludes columns matching: _event, event_, _time, time_, predicted_, _prob, censored, followup, cox_, logistic_. Only numeric predictors are used.

5.13. Kaplan-Meier

Method: Kaplan-Meier survival curves / Nelson-Aalen cumulative hazard

Parameters: | Field | Description | |-----|-----| | `time` | Time-to-event | | `event` | Event indicator | | `group` | Grouping variable | | `stratify` | Stratification variable (separate analysis per stratum) | | `Survival` / `Hazard` | Plot type: survival probability or cumulative hazard | | `95% CI` | Show confidence intervals | | `Chart Axes` | X step (months), Y step (%), labels |

Output metrics:

Metric	Interpretation
<code>summary</code>	Per group: n, events, median survival (months)
Log-rank p	Overall log-rank test p-value. p<0.05 = groups differ significantly
Pairwise log-rank	Bonferroni-corrected pairwise comparisons. Shows which specific groups differ
<code>number_at_risk</code>	Number-at-risk table at key time points
<code>survival_probability</code>	S(t) survival probability table
<code>median_survival</code>	Median survival per group. <code>NR</code> = not reached

Plot: Kaplan-Meier survival curve S(t) or Nelson-Aalen cumulative hazard H(t).

5.14. ROC Analysis

Method: ROC analysis for multiple numeric predictors vs binary target

Parameters: | Field | Description | |-----|-----| | `target` | Binary target | | `Event value` | Value considered "event" (1 or 0) | | `predictors` | Numeric predictors |

Output metrics:

ROC Results tab:

Metric	Interpretation
AUC (95% CI)	Area Under ROC Curve. Hanley-McNeil CI method
Threshold	Optimal cutpoint via Youden's J (sens + spec - 1)
Sensitivity	True positive rate at optimal threshold
Specificity	True negative rate at optimal threshold
PPV	Positive predictive value
NPV	Negative predictive value

Comparison tab (≥ 2 predictors):

Metric	Interpretation
DeLong test	Pairwise AUC comparison. $p < 0.05$ = AUCs differ significantly

Plot: Combined ROC curves for all predictors.

5.15. Model Evaluation (Binary)

Method: Comprehensive binary classification evaluation with bootstrap CI

Parameters: | Field | Description | |-----|-----| | target | Binary target | | predictors | Predicted probability columns (0-1) | | Bootstrap samples | 500, 1000 (default), 2000 | | DCA | Include Decision Curve Analysis | | Calibration | Include calibration curves |

Output metrics:

Metric	Interpretation
AUC (bootstrap CI)	ROC AUC with bootstrap 95% CI
Accuracy	Overall classification accuracy
Sensitivity / Specificity	True positive and negative rates
PPV / NPV	Predictive values
F1	Harmonic mean of precision and recall
Brier Score	Mean squared probability error. 0 = perfect, <0.25 = good, >0.25 = poor
DCA Net Benefit	Clinical net benefit at thresholds 5-50%. $NB = (TP/n) - (FP/n) \times (p/(1-p))$

Plots: - DCA curves (all models on one plot) - Calibration curves (one per model)

5.16. Survival Prediction Evaluation

Method: Survival model evaluation: C-index, time-dependent AUC, bootstrap CI

Parameters: | Field | Description | |-----|-----| | time | Time-to-event | | event | Event indicator | | predictors | Risk score/prediction columns | | Evaluation time points | Time points for AUC (comma-separated, e.g., "12,24,48") | | stratify | Stratification variable | | Bootstrap samples | 500, 1000 (default), 2000 | | Smooth | Smoothed AUC curves |

Output metrics:

Metric	Interpretation
C-index (bootstrap CI)	Harrell's C-index with bootstrap 95% CI. 0.5 = random
Time-dependent AUC	Cumulative dynamic AUC as function of time
ΔC (model comparison)	C-index difference between models with bootstrap p-value
Strata	Separate results per stratum level

Plot: Time-dependent AUC (optionally smoothed).

5.17. Diagnostic Accuracy

Method: Diagnostic accuracy metrics for binary tests vs gold standard

Parameters: | Field | Description | |-----|-----| | `target` |
Reference standard (binary, "gold standard") | | `predictors` |
Index tests (binary or probability 0-1) |

Output metrics (per test):

Metric	Interpretation
Sensitivity (95% CI)	TP/(TP+FN). Proportion of correctly identified positives
Specificity (95% CI)	TN/(TN+FP). Proportion of correctly identified negatives
PPV (95% CI)	TP/(TP+FP). Probability that a positive test reflects true disease
NPV (95% CI)	TN/(TN+FN). Probability that a negative test reflects true absence
Accuracy (95% CI)	(TP+TN)/(Total). Overall correct classification
LR+ (95% CI)	Positive likelihood ratio = Sens/(1-Spec). >10 = strong evidence
LR- (95% CI)	Negative likelihood ratio = (1-Sens)/Spec. <0.1 = strong evidence
DOR	Diagnostic Odds Ratio = (TP×TN)/(FP×FN). 1 = useless test

Comparison tab (>1 test):

Metric	Interpretation
McNemar test	Paired comparison. p<0.05 = tests differ significantly

Contingency tables tab: TP/FP/FN/TN per test.

Plot: Forest plot of sensitivity and specificity with 95% CI.

Note: Probability columns (0-1) are auto-thresholded at 0.5.

5.18. Agreement Analysis

Method: Cohen's Kappa (2 raters) / Weighted Kappa (ordinal) / Fleiss' Kappa (3+ raters)

Parameters: | Field | Description | |-----|-----| |
predictors | Rater columns (≥2 required) | | Nominal /
Ordinal | Scale type | | Linear / Quadratic | Weight type
(ordinal only) |

Output metrics:

Metric	Interpretation
Kappa	Agreement coefficient corrected for chance. Range: -1 to 1
95% CI	Confidence interval
p-value	Significance vs zero
Interpretation	Landis & Koch: Poor (≤ 0), Slight (0.01-0.20), Fair (0.21-0.40), Moderate (0.41-0.60), Substantial (0.61-0.80), Almost Perfect (0.81-1.00)
percent_agreement	Raw agreement percentage

For ≥ 3 raters: Fleiss' Kappa with per-category breakdown.

Plot: Confusion matrix heatmap (2 raters).

5.19. Individual Prediction

Method: Single-patient prediction from saved model with SHAP explanation

Parameters: | Field | Description | |-----|-----| | **Select File** | Upload .joblib or .pkl model | | Patient form | Radio buttons for categorical features, number inputs for numeric |

Output metrics:

Metric	Interpretation
prediction	Predicted class
probability	Probability of class 1

Plot: SHAP waterfall plot — contribution of each feature to the prediction.

6. AI Chat

6.1. AI Settings

Click **"AI Settings"** (left panel or above AI Chat):

Parameter	Description
Provider	Ollama (Local) — run models locally. Groq (Cloud) — cloud API
Ollama URL	Ollama server address (default <code>http://localhost:11434</code>)
Ollama Model	Model (loaded after "Test Connection")
Groq API Key	API key (get at console.groq.com)
Groq Model	Model (list loaded after "Test Connection")
Temperature	Generation temperature: 0 = deterministic, 1 = creative
Max Tokens	Maximum response length (100-8000)
System Prompt	AI instruction prompt (includes project context, dataset info, up to 5 most recent analyses with metrics and output preview)

Buttons: - **"Test Connection"** — test provider connectivity and load model list - **"Connect"** — confirm provider selection - **"Reset to default"** — reset system prompt to default - **Save / Cancel** — save or discard settings

6.2. Modes: Assistant and Coder

AI Assistant (default): - Conversational chat with AI - Answers statistical questions, interprets results - Has context of project description, dataset, and analysis history - Supports markdown formatting

AI Coder ("Coder" toggle): - Generates Python analysis code - Auto-extracts ````python` code blocks and executes them - Shows code output and generated plots - Self-corrects on

errors (up to 3 retries) - Uses temperature 0 (deterministic output)

7. Reports

Right panel → **Reports** tab.

7.1. Standard DOCX Report

Parameter	Description
Analyses	All analyses or Selected (with checkboxes)
Sections	Dataset overview, Results tables, Key metrics
Title	Report title
Generate DOCX	Create report. Download via link

Contents: - Dataset overview (row count, column types) - For each analysis: metrics, tables from preview text

7.2. AI Report

Parameter	Description
Language	Report language (user prompt, default "English")
Generate AI Report	Create AI-generated report

Contents (AI-generated): 1. **Objective** — based on project context 2. **Materials** — dataset description + basic stats 3. **Methods** — statistical methods used 4. **Results** — key findings with exact numbers (HR, OR, p-values, AUC, C-index) 5. **Conclusion** — brief summary

Requirements: - AI must be configured (valid provider and credentials) - At least one analysis in history (max 5 included in prompt)

8. Code Editor Mode

Click **"Code"** in the header.

Action	Description
Type Python code in the monospace text area	
Run	Execute via <code>exec()</code> on the server
Clear Code	Clear editor
Clear History & Plots	Delete all analysis history and plot files
Exit Code Mode	Return to analysis view

Available variables in code: - `df` — project pandas DataFrame - `pd`, `np`, `plt`, `shap` — standard libraries - `save_plot(name)` — save plot with auto-close - `fmt_p(p)` — format p-value (shows `<0.0001` if tiny) - `get_label(var, value)` — variable/value labels

9. Analysis History (Chain)

Right panel → **Chain** tab.

- Chronological list of completed analyses (reverse order, max 200 entries)
- Color coding:
 - **Blue** — regression
 - **Green** — survival
 - **Amber** — evaluation
 - **Gray** — basic tests
 - **Orange** — LASSO
 - **Purple** — Random Forest / Random Survival Forest
- Click an item to restore the analysis card with results

10. Tips & Best Practices

Action	Tip
Enter in AI input	Send message
Shift+Enter	New line
Click a variable	Assign to active field
Hide Charts	Free up memory by hiding plots
 / ☾	Toggle theme

Workflow recommendations

1. **Start with Descriptive Statistics** — inspect distributions, missing values, outliers
2. **Use Categorical + Numeric Comparison** for preliminary screening
3. **For regression:** start with Univariate, then Multivariate (Enter), then Stepwise if needed
4. **Always check Schoenfeld PH test** in Cox regression — this is the key assumption
5. **VIF > 10** — remove collinear predictors and re-run
6. **DCA** helps assess clinical utility, not just statistical significance
7. **Bootstrap CI** provides more realistic metric precision estimates
8. **Don't rely solely on Stepwise p-values** — they don't account for the selection process
9. **LASSO with auto-select C** is a good choice for high-dimensional feature selection
10. **AI Coder mode** is useful for quick ad-hoc analyses without leaving the app